# MODULE 2 – RELATIONS

**Artificial Intelligence and Machine Learning – Semester 3**

Discrete Mathematics

**Definition**

**Syllabus**

- Representation of Relations
- Matrix Representation
- Graph Representation
- Types of Relations
- Properties of Relations
- Equivalence Relations
- Equivalence Classes

## 1. Introduction to Relations

$$A = \{1, 2, 3\}$$

A relation is used to describe a **connection or relationship between elements**.

For example, suppose we want to relate the numbers according to the relation "less than".

Among the elements of

$$A = \{1, 2, 3\},$$

we have

$$1 < 2, \qquad 1 < 3, \qquad 2 < 3.$$

These relationships can be written as ordered pairs:

$$R = \{(1, 2), (1, 3), (2, 3)\}.$$

This is a relation.

**Definition**

**Simple Definition:**
A relation is a set of ordered pairs that describes a relationship between elements.

# 2. Ordered Pair

An ordered pair consists of two elements written in a particular order.
It is written as

$$(a, b).$$

Here,

- $a$ is the first element.
- $b$ is the second element.

**Important**

**The order is important.**

$$(1, 2) \neq (2, 1)$$

because the positions of the elements are different.

## 2.1. Example

Consider

$$(3, 5).$$

The first element is 3 and the second element is 5.
Two ordered pairs are equal if their corresponding elements are equal.

$$(a, b) = (c, d)$$

if and only if

$$a = c$$

and

$$b = d.$$

## 2.2. Example

If

$$(x, 4) = (2, y),$$

then

$$x = 2, \qquad y = 4.$$

# 3. Cartesian Product

Let

$$A = \{1, 2\}$$

and

$$B = \{a, b\}.$$

The Cartesian product of $A$ and $B$ is written as

$$A \times B.$$

It contains all possible ordered pairs where:

- the first element comes from $A$;
- the second element comes from $B$.

Therefore,

$$A \times B = \{(1, a), (1, b), (2, a), (2, b)\}.$$

If

$$|A| = m$$

and

$$|B| = n,$$

then

$$\boxed{|A \times B| = mn}.$$

**Example**

If

$$|A| = 3, \qquad |B| = 4,$$

then

$$|A \times B| = 3 \times 4 = 12.$$

Therefore, $A \times B$ contains 12 ordered pairs.

# 4. Relation

Let $A$ and $B$ be two sets.

A relation from $A$ to $B$ is any subset of $A \times B$.

$$\boxed{R \subseteq A \times B}$$

**Example**

Let

$$A = \{1, 2, 3\}$$

and

$$B = \{a, b\}.$$

Then

$$A \times B = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}.$$

Consider

$$R = \{(1, a), (2, b), (3, a)\}.$$

Since

$$R \subseteq A \times B,$$

$R$ is a relation from $A$ to $B$.

# 5. Relation on a Set

When both sets are the same, we have a relation **on a set**.

For a set $A$, a relation on $A$ is a subset of

$$A \times A.$$

Therefore,

$$\boxed{R \subseteq A \times A}.$$

**Example**

Let

$$A = \{1, 2, 3\}.$$

A relation on $A$ could be

$$R = \{(1, 1), (1, 2), (2, 3)\}.$$

All the ordered pairs belong to $A \times A$.
Therefore, $R$ is a relation on $A$.

# 6. How to Read a Relation

If

$$(a, b) \in R,$$

we say

"$a$ is related to $b$ under relation $R$."

It can also be written as

$$aRb.$$

**Example**

If

$$R = \{(1, 2), (2, 3), (3, 1)\},$$

then

$$(1, 2) \in R$$

means

$$1R2.$$

Therefore, 1 is related to 2.

# 7. Representation of Relations

A relation can be represented in different forms.
For this syllabus, the important representations are:

1. Ordered-pair representation
2. Matrix representation
3. Graph representation

# 8. Matrix Representation of a Relation

A relation can be represented using a matrix containing only 0 and 1.

Suppose

$$A = \{1, 2, 3\}$$

and

$$R = \{(1, 1), (1, 2), (2, 3), (3, 1)\}.$$

To construct the matrix:

- First element of the ordered pair $\rightarrow$ Row
- Second element of the ordered pair $\rightarrow$ Column
- Pair present in $R \rightarrow 1$
- Pair absent from $R \rightarrow 0$

Therefore,

| $R$ | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 1 | 1 | 0 |
| 2 | 0 | 0 | 1 |
| 3 | 1 | 0 | 0 |

Hence,

$$\boxed{M_R = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}}$$

**Important**

$\boxed{\text{First element} = \text{Row}}$

$\boxed{\text{Second element} = \text{Column}}$

$\boxed{\text{Pair present} = 1}$

$\boxed{\text{Pair absent} = 0}$

# 9. Graph Representation of a Relation

A relation on a set can also be represented using a directed graph.

In a directed graph:

- Elements of the set are represented by vertices.
- An ordered pair $(a, b)$ is represented by an arrow from $a$ to $b$.
- If $(a, a) \in R$, a loop is drawn at $a$.

**Example**

Let

$$A = \{1, 2, 3\}$$

and

$$R = \{(1, 2), (2, 3), (3, 1)\}.$$

The graph contains the directed arrows

$$1 \to 2,$$

$$2 \to 3,$$

and

$$3 \to 1.$$

### 9.1. Loop

If

$$(a, a) \in R,$$

we draw an arrow from $a$ back to itself.
This is called a **loop**.
For example,

$$(1, 1) \in R$$

is represented by a loop at vertex 1.

## 10. Example: Relation Defined by $x > y$

This type of question is important for examinations.
Let

$$X = \{1, 2, 3, 4\}$$

and

$$R = \{(x, y) \mid x > y\}.$$

We need to find all ordered pairs satisfying $x > y$.

### For $x = 1$

There is no element in $X$ smaller than 1.
Therefore, there is no pair.

### For $x = 2$

$$2 > 1.$$

Therefore,

$$(2, 1)$$

belongs to $R$.

### For $x = 3$

$$3 > 1, \qquad 3 > 2.$$

Therefore,

$$(3, 1), (3, 2)$$

belong to $R$.

### For $x = 4$

$$4 > 1, \qquad 4 > 2, \qquad 4 > 3.$$

Therefore,

$$(4, 1), (4, 2), (4, 3)$$

belong to $R$.
Hence,

$$\boxed{R = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}}$$

## 10.1. Matrix Representation

Rows and columns are arranged in the order

$$1, 2, 3, 4.$$

The matrix is

$$\boxed{M_R = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}}$$

because, for example,

$$(2,1) \in R$$

so row 2, column 1 contains 1.
Similarly,

$$(3,1),(3,2) \in R$$

so row 3 contains

$$1 \quad 1 \quad 0 \quad 0.$$

### 10.2. Graph Representation

The arrows are

$$2 \to 1,$$

$$3 \to 1, \qquad 3 \to 2,$$

$$4 \to 1, \qquad 4 \to 2, \qquad 4 \to 3.$$

There are no loops because

$$x > x$$

is never true.

## 11. Types of Relations

There are several special types of relations.
The basic types are:

1. Empty relation
2. Universal relation
3. Identity relation

Properties such as reflexive, symmetric, antisymmetric and transitive are discussed separately.

## 11.1. Empty Relation

A relation containing no ordered pairs is called an **empty relation**.

$$\boxed{R = \varnothing}$$

**Example**

Let

$$A = \{1, 2, 3\}.$$

If

$$R = \varnothing,$$

then no element is related to any element.

## 11.2. Universal Relation

A relation containing every possible ordered pair is called a **universal relation**.

$$\boxed{R = A \times A}$$

**Example**

Let

$$A = \{1, 2\}.$$

Then

$$A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}.$$

Therefore, the universal relation is

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2)\}.$$

## 11.3. Identity Relation

The identity relation contains only pairs in which both elements are the same.

$$\boxed{I_A = \{(a, a) : a \in A\}}$$

**Example**

For

$$A = \{1, 2, 3\},$$

the identity relation is

$$\boxed{I_A = \{(1, 1), (2, 2), (3, 3)\}.}$$

Every element is related to itself.

# 12. Properties of Relations

The important properties of relations are:

1. Reflexive
2. Irreflexive
3. Symmetric
4. Antisymmetric
5. Transitive

# 13. Reflexive Relation

A relation $R$ on $A$ is called **reflexive** if every element is related to itself.
Mathematically,

$$\boxed{(a, a) \in R \quad \forall a \in A}$$

**Example**

Let

$$A = \{1, 2, 3\}$$

and

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}.$$

Since

$$(1, 1), (2, 2), (3, 3) \in R,$$

the relation is reflexive.

**Important**

**Easy way to remember:**
Reflexive = Every element is related to itself.

# 14. Irreflexive Relation

A relation is called **irreflexive** if no element is related to itself.

$$\boxed{(a,a) \notin R \quad \forall a \in A}$$

**Example**

Let

$$A = \{1, 2, 3\}$$

and

$$R = \{(1,2), (2,3), (3,1)\}.$$

There are no self-pairs:

$$(1,1), (2,2), (3,3).$$

Therefore, $R$ is irreflexive.

# 15. Symmetric Relation

A relation is called **symmetric** if

$$\boxed{(a,b) \in R \Rightarrow (b,a) \in R}.$$

In simple words:

If $a$ is related to $b$, then $b$ must also be related to $a$.

**Example**

Consider

$$R = \{(1,2),(2,1),(2,3),(3,2)\}.$$

We have

$$(1,2) \in R$$

and

$$(2,1) \in R.$$

Also,

$$(2,3) \in R$$

and

$$(3,2) \in R.$$

Therefore, the relation is symmetric.

**Important**

$$\boxed{a \to b \quad \Rightarrow \quad b \to a}$$

This is a simple way to remember a symmetric relation.

# 16. Antisymmetric Relation

A relation is called **antisymmetric** if

$$(a,b) \in R$$

and

$$(b,a) \in R$$

imply

$$\boxed{a = b}.$$

In simple words:

Two different elements cannot be related to each other in both directions.

**Example**

Let

$$A = \{1, 2, 3\}$$

and

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}.$$

We have

$$(1, 2) \in R$$

but

$$(2, 1) \notin R.$$

Therefore, there is no violation of antisymmetry.
Hence, the relation is antisymmetric.

## 16.1. Symmetric vs Antisymmetric

This is a common source of confusion.

### Symmetric

If

$$1R2,$$

then we must have

$$2R1.$$

### Antisymmetric

If both

$$1R2$$

and

$$2R1$$

are present, then we must have

$$1 = 2.$$

Since

$$1 \neq 2,$$

both directions cannot occur for two different elements in an antisymmetric relation.

**Important**

**Antisymmetric does not mean "not symmetric".**
A relation can be both symmetric and antisymmetric.
For example, the identity relation

$$R = \{(1,1),(2,2),(3,3)\}$$

is both symmetric and antisymmetric.

# 17. Transitive Relation

A relation is called **transitive** if

$$\boxed{(a,b) \in R,\ (b,c) \in R \Rightarrow (a,c) \in R}$$

In simple words:

If $a$ is related to $b$, and $b$ is related to $c$, then $a$ must be related to $c$.

**Example**

Consider

$$R = \{(1,2),(2,3),(1,3)\}.$$

We have

$$(1,2) \in R$$

and

$$(2,3) \in R.$$

Therefore, transitivity requires

$$(1,3) \in R.$$

It is present.
Therefore, this relation satisfies the transitive requirement.

**Important**

Think of transitivity as

$$1 \to 2, \qquad 2 \to 3$$

requiring

$$1 \to 3.$$

# 18. Quick Comparison of Properties

| **Property** | Main Condition |
|---|---|
| **Reflexive** | $(a, a)$ must be present for every $a \in A$ |
| **Irreflexive** | $(a, a)$ must not be present for any $a \in A$ |
| **Symmetric** | $(a, b) \in R \Rightarrow (b, a) \in R$ |
| **Antisymmetric** | $(a, b), (b, a) \in R \Rightarrow a = b$ |
| **Transitive** | $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$ |

# 19. How to Check Properties in an Exam

Suppose

$$A = \{1, 2, 3\}$$

and

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}.$$

We can check the properties one by one.

## Step 1: Reflexive

Check:

$$(1, 1), (2, 2), (3, 3).$$

All are present.
Therefore,

$$\boxed{\text{Reflexive}}$$

## Step 2: Symmetric

We have

$$(1, 2) \in R$$

and

$$(2, 1) \in R.$$

Therefore,

$$\boxed{\text{Symmetric}}$$

### Step 3: Antisymmetric

We have both

$$(1,2) \in R$$

and

$$(2,1) \in R,$$

but

$$1 \neq 2.$$

Therefore,

$$\boxed{\text{Not antisymmetric}}$$

### Step 4: Transitive

For example,

$$(1,2), (2,1) \in R.$$

Transitivity requires

$$(1,1) \in R,$$

which is present.
Similarly,

$$(2,1), (1,2) \in R$$

requires

$$(2,2) \in R,$$

which is also present.
Therefore,

$$\boxed{\text{Transitive}}$$

## 20. Previous-Year Type Question

**Exam Point**

**Question:**
If

$$A = \{1,2,3,4\},$$

give an example of a relation on $A$ which is reflexive and transitive, but not symmetric.

One suitable relation is defined using the relation "less than or equal to":

$$\boxed{R = \{(x, y) \in A \times A : x \leq y\}}$$

Writing all ordered pairs,

$$\begin{aligned} R = \{&(1,1), (1,2), (1,3), (1,4), \\ &(2,2), (2,3), (2,4), \\ &(3,3), (3,4), \\ &(4,4)\}. \end{aligned}$$

## Why is it reflexive?

For every element,

$$1 \leq 1, \quad 2 \leq 2, \quad 3 \leq 3, \quad 4 \leq 4.$$

Therefore,

$$(1,1), (2,2), (3,3), (4,4) \in R.$$

Hence, $R$ is reflexive.

## Why is it transitive?

If

$$a \leq b$$

and

$$b \leq c,$$

then

$$a \leq c.$$

Therefore, $R$ is transitive.

## Why is it not symmetric?

We have

$$1 \leq 2,$$

so

$$(1,2) \in R.$$

But

$$2 \leq 1$$

is false.
Therefore,

$$(2,1) \notin R.$$

Hence, $R$ is not symmetric.
Therefore,

$$\boxed{R \text{ is reflexive and transitive, but not symmetric.}}$$

# 21. Equivalence Relation

An equivalence relation is a special type of relation.

A relation $R$ is an equivalence relation if it satisfies the following three properties:

1. Reflexive
2. Symmetric
3. Transitive

Therefore,

$$\boxed{\text{Equivalence Relation} = \text{Reflexive} + \text{Symmetric} + \text{Transitive}}$$

**Important**

A useful memory aid is:

$$\boxed{\text{RST}}$$

- R – Reflexive
- S – Symmetric
- T – Transitive

**Antisymmetric is not required for an equivalence relation.**

# 22. Example of an Equivalence Relation

Let

$$A = \{1, 2, 3, 4\}$$

and

$$R = \{(1,1), (1,3), (3,1), (3,3), (2,2), (2,4), (4,2), (4,4)\}.$$

We check the three required properties.

### Reflexive

The pairs

$$(1,1),(2,2),(3,3),(4,4)$$

are present.
Therefore, $R$ is reflexive.

### Symmetric

We have

$$(1,3) \text{ and } (3,1)$$

and

$$(2,4) \text{ and } (4,2).$$

Therefore, $R$ is symmetric.

### Transitive

The elements form two groups:

$$\{1,3\}$$

and

$$\{2,4\}.$$

All required relationships within each group are present.
Therefore, $R$ is transitive.
Hence,

$$\boxed{R \text{ is an equivalence relation.}}$$

## 23. How to Identify an Equivalence Relation

If an examination question asks:

> Determine whether $R$ is an equivalence relation.

Students should follow these steps.

### Step 1

Show that $R$ is reflexive.

### Step 2

Show that $R$ is symmetric.

### Step 3

Show that $R$ is transitive.

### Step 4

Write the conclusion:

$$\boxed{R \text{ is an equivalence relation.}}$$

If even one of the three required properties fails, then:

$$\boxed{R \text{ is not an equivalence relation.}}$$

## 24. Equivalence Class

Once we have an equivalence relation, we can group related elements.

The equivalence class of an element $a$ is the set of all elements that are related to $a$.

It is written as

$$\boxed{[a] = \{x \in A : xRa\}}$$

In simple words:

> The equivalence class of $a$ contains all elements of $A$ that are related to $a$.

## 25. Finding Equivalence Classes

Consider

$$R = \{(1,1), (1,3), (3,1), (3,3), (2,2), (2,4), (4,2), (4,4)\}.$$

### Find $[1]$

Look for all elements related to 1.

The relevant pairs are

$$(1,1), (3,1).$$

Therefore,

$$\boxed{[1] = \{1,3\}}$$

### Find [3]

The relevant pairs are

$$(1,3),(3,3).$$

Therefore,

$$\boxed{[3]=\{1,3\}}$$

Hence,

$$[1]=[3].$$

### Find [2]

The relevant pairs are

$$(2,2),(4,2).$$

Therefore,

$$\boxed{[2]=\{2,4\}}$$

Similarly,

$$\boxed{[4]=\{2,4\}}$$

Therefore, the distinct equivalence classes are

$$\boxed{\{1,3\},\{2,4\}}.$$

## 26. Understanding Equivalence Classes

An equivalence relation divides the elements of a set into groups.

For example,

$$A=\{1,2,3,4\}$$

is divided into

$$\{1,3\}$$

and

$$\{2,4\}.$$

Elements belonging to the same group are related to one another.

Thus,

$$1R3$$

and

$$2R4.$$

But,

$$1 \not R 2.$$

The groups formed by an equivalence relation are called **equivalence classes**.

# 27. Complete Exam-Type Question

**Exam Point**

Let

$$A = \{1, 2, 3, 4\}$$

and

$$R = \{(1,1), (1,3), (3,1), (3,3), (2,2), (2,4), (4,2), (4,4)\}.$$

1. Determine whether $R$ is an equivalence relation.
2. Find the equivalence classes.

## Solution

### Step 1: Reflexive

We have

$$(1,1), (2,2), (3,3), (4,4) \in R.$$

Therefore, $R$ is reflexive.

### Step 2: Symmetric

The reverse pairs are also present:

$$(1,3) \leftrightarrow (3,1)$$

and

$$(2,4) \leftrightarrow (4,2).$$

Therefore, $R$ is symmetric.

**Step 3: Transitive**

The elements form the groups

$$\{1, 3\}$$

and

$$\{2, 4\}.$$

All required relationships within the groups are present.
Therefore, $R$ is transitive.
Hence,

$$\boxed{R \text{ is an equivalence relation.}}$$

**Equivalence Classes**

$$\boxed{[1] = [3] = \{1, 3\}}$$

and

$$\boxed{[2] = [4] = \{2, 4\}}.$$

Therefore, the distinct equivalence classes are

$$\boxed{\{1, 3\}, \{2, 4\}}.$$

# 28. Previous-Year Question: Find All Subsets

This question belongs to Set Theory, but students should already know this concept.

**Exam Point**

Find all subsets of the set

$$\{1, 2, 3\}.$$

Let

$$A = \{1, 2, 3\}.$$

The subsets are:

## Empty subset

$$\varnothing$$

## One-element subsets

$$\{1\}, \quad \{2\}, \quad \{3\}$$

### Two-element subsets

$$\{1,2\}, \quad \{1,3\}, \quad \{2,3\}$$

### Three-element subset

$$\{1,2,3\}$$

Therefore,

$$\boxed{\mathcal{P}(A) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}}$$

Since $A$ has 3 elements,

$$\boxed{\text{Number of subsets} = 2^3 = 8}.$$

## 29. Previous-Year Question Patterns

Students should be able to solve the following types of questions after completing this topic.

### Type 1: Construct a Relation

Give an example of a relation on a set which is:

- reflexive and transitive but not symmetric;
- symmetric;
- reflexive;
- antisymmetric;
- transitive.

For example,

$$R = \{(x,y) \in A \times A : x \leq y\}$$

is reflexive and transitive but not symmetric.

### Type 2: Find a Relation from a Condition

Given

$$X = \{1,2,3,4\}$$

and

$$R = \{(x,y) : x > y\},$$

students should list all ordered pairs satisfying the condition.

## Type 3: Matrix Representation

Students should convert a relation into a matrix using

$$\boxed{\text{Pair present} \to 1}$$

and

$$\boxed{\text{Pair absent} \to 0}.$$

## Type 4: Graph Representation

Students should convert each ordered pair

$$(a, b)$$

into a directed arrow

$$a \to b.$$

## Type 5: Explain Types of Relations

Students should be able to define and give examples of:

- Empty relation
- Universal relation
- Identity relation
- Reflexive relation
- Irreflexive relation
- Symmetric relation
- Antisymmetric relation
- Transitive relation

## Type 6: Identify an Equivalence Relation

Given a relation, students should check:

$$\boxed{\text{Reflexive} + \text{Symmetric} + \text{Transitive}}$$

and decide whether it is an equivalence relation.

## Type 7: Find Equivalence Classes

Given an equivalence relation, students should find

$$[1], [2], [3], \ldots$$

and identify the distinct equivalence classes.

# 30. Quick Revision Table

| **Concept** | What to Remember |
|---|---|
| **Ordered Pair** | $(a, b)$; order matters. |
| **Cartesian Product** | $A \times B$ contains all possible ordered pairs. |
| **Relation** | $R \subseteq A \times B$. |
| **Relation on $A$** | $R \subseteq A \times A$. |
| **Matrix** | Pair present $\rightarrow 1$, pair absent $\rightarrow 0$. |
| **Graph** | $(a, b) \rightarrow a \rightarrow b$. |
| **Empty Relation** | $R = \varnothing$. |
| **Universal Relation** | $R = A \times A$. |
| **Identity Relation** | $I_A = \{(a, a) : a \in A\}$. |
| **Reflexive** | Every $a$ has $(a, a)$. |
| **Irreflexive** | No $(a, a)$ is present. |
| **Symmetric** | $(a, b) \Rightarrow (b, a)$. |
| **Antisymmetric** | $(a, b), (b, a) \Rightarrow a = b$. |
| **Transitive** | $(a, b), (b, c) \Rightarrow (a, c)$. |
| **Equivalence Relation** | Reflexive + Symmetric + Transitive. |
| **Equivalence Class** | $[a] = \{x \in A : xRa\}$. |

# 31. Important Exam Reminders

**Important**

1. A relation from $A$ to $B$ is a subset of $A \times B$.
2. A relation on $A$ is a subset of $A \times A$.
3. In a matrix:

$$\boxed{\text{First element} = \text{Row, Second element} = \text{Column}}$$

4. In a graph:

$$\boxed{(a, b) \to a \to b}$$

5. Reflexive means every element has a self-pair.
6. Symmetric means every pair has its reverse pair.
7. Antisymmetric means two different elements cannot have pairs in both directions.
8. Transitive means:

$$(a, b), (b, c) \Rightarrow (a, c).$$

9. An equivalence relation must be:

$$\boxed{\text{Reflexive, Symmetric and Transitive}}$$

10. Equivalence classes contain elements that are related to each other.

# 32. Concept Flow

$$\boxed{\text{Sets} \to \text{Ordered Pairs} \to \text{Cartesian Product} \to \text{Relation}}$$

$$\boxed{\text{Relation} \to \begin{cases} \text{Matrix Representation} \\ \text{Graph Representation} \\ \text{Properties} \end{cases}}$$

$$\boxed{\text{Reflexive} + \text{Symmetric} + \text{Transitive} \to \text{Equivalence Relation} \to \text{Equivalence Classes}}$$

# 33. Practice Questions

1. Let

$$A = \{1, 2, 3\}$$

and

$$B = \{a, b\}.$$

Find $A \times B$.

2. Define a relation from $A$ to $B$.

3. Let

$$A = \{1, 2, 3\}$$

and

$$R = \{(1, 1), (1, 2), (2, 3)\}.$$

Find the matrix representation of $R$.

4. Represent

$$R = \{(1, 2), (2, 3), (3, 1)\}$$

using a directed graph.

5. Let

$$A = \{1, 2, 3, 4\}$$

and

$$R = \{(x, y) : x > y\}.$$

Find $R$ and its matrix.

6. Give an example of an empty relation.

7. Give an example of a universal relation on

$$A = \{1, 2\}.$$

8. Give an example of an identity relation on

$$A = \{1, 2, 3\}.$$

9. Define reflexive, symmetric and transitive relations with examples.

10. If

$$A = \{1, 2, 3, 4\},$$

give an example of a relation which is reflexive and transitive but not symmetric.

11. Determine whether the following relation is an equivalence relation:

$$R = \{(1, 1), (1, 3), (3, 1), (3, 3), (2, 2), (2, 4), (4, 2), (4, 4)\}.$$

12. Find the equivalence classes of the relation in Question 11.

# 34. Summary

A relation describes a connection between elements using ordered pairs.

The important sequence to remember is:

$$\boxed{A, B \to A \times B \to R \subseteq A \times B}$$

A relation can be represented using:

$$\boxed{\text{Ordered pairs, Matrix and Graph}}$$

The important properties are:

$$\boxed{\text{Reflexive, Irreflexive, Symmetric, Antisymmetric, Transitive}}$$

An equivalence relation satisfies:

$$\boxed{\text{Reflexive + Symmetric + Transitive}}$$

and the groups formed by an equivalence relation are called:

$$\boxed{\text{Equivalence Classes}}$$